

Somnath Pandit

Full Stack & IoT Software Engineer

+91 8967650078 ✉ somnathpandit633@gmail.com in somnath-pandit GitHub somnath503 📍 West Bengal, India

SUMMARY

Full Stack & IoT Software Engineer with experience in developing web applications, IoT solutions, and real-time data pipelines. Proficient in Node.js, C++, React, Spring Boot, and embedded systems. Passionate about building scalable software, microservices, and end-to-end applications that connect hardware devices with modern web platforms.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Nov 2022 – Jun 2026

University of Kalyani

Kalyani, India

- **Leadership:** Academic General Secretary (AGS), Department of CSE (2024)

SKILLS

Languages: Java 17, TypeScript/JavaScript, C/C++, Python, SQL

Backend & Frameworks: Node.js, Express.js, Spring Boot, FastAPI, C++ (Crow, Boost.Asio), REST APIs, MQTT, WebSockets

Frontend: React 19, Next.js, ReactFlow, Vite, Tailwind CSS, Material UI, Android

Databases & Storage: MySQL, PostgreSQL (Native pg), SQLite

Systems & Infrastructure: Linux (Ubuntu 24.04 LTS), Nginx, PM2, Docker, ESP32 Dev Kit V1, GSM EC200, Git

EXPERIENCE

Full Stack Engineer Intern

12/2025 – 06/2026

Lilac Inc

Kolkata, India

- **Audlog Pro Voicelogger:** Built a production audio logging server (Node.js, React, MySQL) on Ubuntu. Configured Nginx, PM2, and custom filesystem pipelines for large-scale .opus storage.
- **IoT Telemetry & LC Gate:** Engineered a real-time railway monitoring system using the ESP32 Dev Kit V1, MQTT, and WebSockets. Integrated EC200 GSM modules for remote web-based network management.
- **Hardware Metrics (C++):** Developed Linux microservices via C++ (Crow, Boost.Asio) to expose CPU, disk, and network states through REST APIs.

PROJECTS

Visual DAG Execution Platform | React, ReactFlow, FastAPI, Python

Live | GitHub

- Built a visual node-based editor for designing and validating complex AI data pipelines directly in the browser.
- Implemented Kahn's Topological Sort Algorithm on the FastAPI backend to detect graph cycles and validate Directed Acyclic Graphs (DAGs) in $O(V + E)$ time.

NexHire ATS Ecosystem | React 19, Node.js, PostgreSQL, Zod

Live | GitHub

- Engineered a production-ready Applicant Tracking System featuring a decoupled architecture separating client-side rendering from backend data APIs.
- Utilized direct PostgreSQL queries via the native pg client for zero-ORM performance overhead, enforcing strict schema validation with Zod and JWT role-based access control.

GA - Scheduler (NP-Hard Solver) | Next.js, Zustand, Vite PWA

Live | GitHub

- Authored an offline-first Progressive Web App to solve the University Course Timetabling Problem entirely client-side.
- Designed a matrix-centric Genetic Algorithm avoiding 1D chromosome destruction, achieving 42% faster convergence through targeted heuristic mutation and vectorized $O(1)$ constraint lookups.

dev.restaurant Ecosystem | Spring Boot, React, Android, Microservices

Live | GitHub

- Architected a highly scalable 4-tier microservice system for seamless food delivery operations.
- Integrated web dashboards and mobile touchpoints to synchronize real-time order processing and restaurant management.

ZeroProof: Anonymous Feedback System | Node.js, TypeScript, React

Live | GitHub

- Architected an anonymous feedback system using RSA Blind Signatures to cryptographically decouple voter identities from submission receipts.
- Engineered OTP invite workflows, token issuance, and admin verification dashboards via secure REST APIs.